

Tan Dai Ngo

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EDUCATION

University of Victoria | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Courses:** Algorithms and Data Models, Systems for Massive Datasets, Information Theory. **GPA:** 8.78 / 9.00.

University of Chicago | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Courses:** Distributed Systems, Databases, Applied Software Engineering, Networks.

University of Washington | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00.

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot, keeping Cassandra query response times under 3 seconds.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Debugged production issues with Splunk and built a Mobileum email notifier over the Microsoft Graph API.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering fundamentals to 50+ students.
- Led coding projects focused on debugging, algorithmic reasoning and performance optimization.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, PyTorch, XGBoost *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six classifiers on 206,000 CIC IoT-DIAD 2024 flows. XGBoost reached 99.5% recall at 0.94% FPR.
- Designed a four condition holdout that exposed session leakage, dropping PR-AUC from 0.981 to 0.125.
- Scripted an integrity check rebuilding the 60 second connection count on a truncated capture, proving causality.

Ocean Data ML Platform with RAG | Python, Spark, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Built Bronze and Silver data layers from 10.8 million NOAA and Ocean Networks Canada observations, then produced a Gold forecasting table with 3.69 million rows.
- Served model inference and plain English retrieval behind FastAPI with Qdrant and Ollama, at hit@k 0.90.
- Tuned XGBoost with Optuna and tracked runs in MLflow. RMSE fell 48% against persistence twelve hours ahead.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises over rolling 60 minute windows, with per market CSV thresholds, cooldowns and trade caps.
- Built a second strategy that bought only when the EMA over 8 periods was above the EMA over 20, price was above the EMA over 200 periods and the trade stayed positive after fees.
- Backtested both strategies on four months of Coinbase data and deployed to EC2 with Lambda and SNS alerts.

TECHNICAL SKILLS

Languages & Programming: Python, Java, object oriented programming, scripting

Machine Learning & Evaluation: PyTorch, XGBoost, scikit-learn, pandas, NumPy, Optuna, MLflow, classification, time-series forecasting, model validation

Data, Retrieval & Messaging: Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

Backend, Cloud & CI/CD: Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda, SNS), Jenkins deployments, Git, Splunk production debugging and monitoring

Engineering Practice: pytest, API testing, model evaluation, experiment tracking, code review, Postman, Jira